

REYNARD SETIAWAN

BSD City, South Tangerang, Indonesia • +62 895 3519 69010 • reynardsetiawan.rs@gmail.com • linkedin.com/in/reynardsetiawan- • github.com/Reynardstw

ABOUT ME

Final-year Computer Science student at Bina Nusantara University, with hands-on experience building mobile applications, AI integration, and embedded systems. Proficient in frontend development, machine learning, and deep learning, with a growing foundation in full-stack development. Comfortable working in teams, stepping up to lead, taking full ownership of my part, and solving problems with a systematic approach.

WORKING EXPERIENCE

Apple Developer Academy @BINUS, Tangerang

March – December 2026 (Expected)

iOS Developer

- Developed user-centered mobile applications through Challenge-Based Learning, working within a team to design and build technical solutions for real-world problems.

EDUCATION

BINA NUSANTARA UNIVERSITY – Bandung, Indonesia

2023 — 2027 (Expected)

Undergraduate Computer Science, Data Science Stream • Bandung, Indonesia

- GPA: 3.86/4.00
- Relevant Coursework: Software Engineering, Machine Learning, Deep Learning, Human-Computer Interaction, Data Mining, Data Structures, Database Systems, Object-Oriented Programming.

PROJECTS

Full portfolio: [PORTFOLIO](#)

VoxA — Audio Recording, Transcription & Translation App

May — Jun 2025

Frontend Developer & AI Integration • Flutter • github.com/Reynardstw/VoxA

- Focused on a **clean, intuitive interface for recording, transcribing, and translating audio**, minimizing friction across the full recording-to-translation flow.
- Integrated a self-deployed **Whisper** model for transcription and **DistilBART** for summarization, connecting the app to **custom AI endpoints**.
- Collaborated in a **3-person team**, owning the **frontend and AI integration layers end-to-end from prototype** to working app.

Iyup — Shade-Finding App with ML Forecasting & IoT Integration

July – August 2026

iOS (Swift, SwiftUI) • github.com/Reynardstw/Iyup

- Developed **6 custom Core ML models** to forecast shade and environmental conditions, integrated directly into the iOS app.
- Implemented **RealityKit** integration and the data-receiving side of a real-time IoT pipeline from an **ESP32 device** via **MQTT**.
- Structured the **project with a per-feature modular MVVM architecture** and protocol-based dependency injection, working in a **5-person team** with structured Git workflows.

Petik.in — Guitar Chord Progression Recognition App

April – May 2026

iOS (Swift, SwiftUI) • github.com/CicakBelanda/Petik.in-Guitar-Chord-Progression

- Designed a **modular audio processing pipeline** (input, buffering, DSP feature extraction, ML inference, validation) to detect chords in **real time** from live audio.
- **Refactored code** across the shared codebase and **resolved recurring merge conflicts**, keeping the team's Git workflow unblocked.
- Collaborated in a **5-person team** with clearly separated modules (audio, camera-based chord detection, UI), enabling parallel development.

Orange Sorting System — Embedded Firmware for Automated Fruit Sorting

July – August 2026

Embedded Swift, ESP-IDF • ESP32-C6

- Implemented sensor logic in Embedded Swift, including load cell weight reading (**HX711**) with auto-tare and stability detection.
- Built the **MQTT** communication layer to relay real-time sorting data from the device to the rest of the system.
- Worked hands-on across the hardware-software boundary, from low-level GPIO/timing logic to networked data output.

PUBLICATIONS

Representation-based Anomaly Detection with Adaptive Realignment for Cross-Domain NIDS

2026

1st Author • 2026 IEEE International Conference on Instrumentation, Networks, Future Sustainability, Information Theory, Transportation, and Systems (INFINITYS), Solo, Indonesia

- Proposed a cross-domain network anomaly detection framework using a **Variational Autoencoder (VAE)** and **RealINVP** normalizing flow; achieved up to **0.995 AUROC** on the NetFlow benchmark, outperforming other baselines.
- Designed the **cross-domain dataset merging** approach and experimental methodology, **benchmarked** against baseline models, and validated model performance through **multi-seed** testing.

Optimizing Disease Diagnosis Accuracy Through Machine Learning

2025

Co-Author • 2025 IEEE International Conference on Internet of Things and Intelligence Systems (IoTais)

- Developed a stacked ensemble model (**CatBoost, XGBoost, LightGBM**) for disease diagnosis, evaluated across 7 medical datasets with **0.885–0.963 accuracy**.
- Co-designed the **ensemble** combination of the three models, sourced the medical datasets from public sources, and presented the research findings online.

ORGANIZATION

Independence Day Committee — Apple Developer Academy @BINUS

August 2026

Logistics & Event Preparation Committee

- Planned and determined equipment and supply needs for the event, ensuring every requirement was accounted for and ready ahead of the event day.
- Distributed equipment and supplies to each event PIC, coordinating within the team under tight deadlines.
- Helped ensure the event ran systematically, keeping each part of the program on track with the plan.

Polling Station Committee (KPPS)

February & November 2024

Member • 2024 General Election & Regional Election (Indonesia)

- Handled ballot distribution, voter document verification, and vote counting on election day.
- Worked closely with team members under the observation of witnesses and voters, maintaining accuracy and precision throughout a long and high-pressure process.

COMPETITIONS

Find IT — Universitas Gadjah Mada

2025

Participant • Competitive Programming

- Competed in a national-level programming competition as part of a 3-person team, solving competitive programming problems under time constraints.
- Solved algorithmic problems using **C++** with the team, dividing problem-solving strategy effectively.

Technoscape 8.0 - BNCC

2025

Group Contributor: Ideation & System Planning • Hackathon (Conceptual Prototype)

- Conceptualized GoChain, a web-based platform addressing claim delays, fraud, and lack of transparency in Indonesia's BPJS Kesehatan health insurance claim process, integrating AI for fraud detection and blockchain for data transparency.
- Contributed to system flow design, stakeholder role mapping, and SWOT and Business Model Canvas analysis for the final proposal.

SKILLS

Mobile: Swift, SwiftUI, Flutter, Dart

AI/ML: Core ML, Model Deployment (HuggingFace Spaces/Inference API), Python, Machine Learning, Deep Learning, Ensemble Learning

Embedded/IoT: ESP32, MQTT

Full-Stack: React, TypeScript, Node.js/Express, PostgreSQL/MongoDB, Prisma, Jest

Tools: REST API Integration, Git/GitHub, Xcode, Docker

Languages: Indonesia (Native), English (Fluent)